

Lecture 5 – Identifiers (2)

COSE212: Programming Languages

Jihyeok Park



2026 Fall

- **Identifiers**
 - Bound identifiers
 - Free identifiers
 - Shadowing
- VAE – AE **with variables**
 - Concrete syntax
 - Abstract syntax

- **Identifiers**
 - Bound identifiers
 - Free identifiers
 - Shadowing
- VAE – AE **with variables**
 - Concrete syntax
 - Abstract syntax
- In this lecture, we will
 - implement the **interpreter** for VAE
 - define the **natural semantics** for VAE

1. Evaluation with Environments

2. Interpreter and Natural Semantics for VAE

- Numbers

- Addition and Multiplication

- Variable Definition

- Variable Lookup

3. Examples

1. Evaluation with Environments

2. Interpreter and Natural Semantics for VAE

Numbers

Addition and Multiplication

Variable Definition

Variable Lookup

3. Examples

Let's evaluate the following VAE expressions:

```
/* VAE */  
val x = 1; {           // [ x -> 1 ]  
  val y = 2; {         // [ x -> 1, y -> 2 ]  
    x + y              // x + y = 1 + 2 = 3  
  }  
}
```

Let's evaluate the following VAE expressions:

```
/* VAE */  
val x = 1; {           // [ x -> 1 ]  
  val y = 2; {         // [ x -> 1, y -> 2 ]  
    x + y              // x + y = 1 + 2 = 3  
  }  
}
```

How to evaluate the expression $x + y$ into the value 3?

$$\vdash x + y \Rightarrow 3$$

Let's evaluate the following VAE expressions:

```
/* VAE */  
val x = 1; {           // [ x -> 1 ]  
  val y = 2; {         // [ x -> 1, y -> 2 ]  
    x + y              // x + y = 1 + 2 = 3  
  }  
}
```

How to evaluate the expression $x + y$ into the value 3?

$$\vdash x + y \Rightarrow 3$$

We need to keep track of the **environment** that maps identifiers to values:

$$[x \mapsto 1, y \mapsto 2] \vdash x + y \Rightarrow 3$$

```
type Value = BigInt           // values
def interp(expr: Expr): Value = ... // interpreter
```

For AE, the interpreter takes an expression and returns a number.

$$\vdash e \Rightarrow n$$

```
type Value = BigInt                                // values
type Env = Map[String, Value]                      // environments
def interp(expr: Expr, env: Env): Value = ... // interpreter
```

For VAE, we extend the interpreter to take an **environment** as well.

$$\sigma \vdash e \Rightarrow n$$

We read it as “*with the **environment** σ , the **expression** e evaluates to the **number** n* ”

```
type Value = BigInt                                // values
type Env = Map[String, Value]                     // environments
def interp(expr: Expr, env: Env): Value = ...    // interpreter
```

For VAE, we extend the interpreter to take an **environment** as well.

$$\sigma \vdash e \Rightarrow n$$

We read it as “*with the **environment** σ , the **expression** e evaluates to the **number** n* ”

For example, the interpreter should be able to evaluate like this:

```
val env : Env    = Map("x" -> 1, "y" -> 2)    // [ x -> 1, y -> 2 ]
val expr: Expr   = Expr("x + y")              // Add(Id("x"), Id("y"))
val v    : Value = interp(expr, env)           // 3
```

$$[x \mapsto 1, y \mapsto 2] \vdash x + y \Rightarrow 3$$

1. Evaluation with Environments

2. Interpreter and Natural Semantics for VAE

- Numbers

- Addition and Multiplication

- Variable Definition

- Variable Lookup

3. Examples

For VAE, we need to 1) implement the **interpreter** with **environments**

```
def interp(expr: Expr, env: Env): Value = ???
```

For VAE, we need to 1) implement the **interpreter** with **environments**

```
def interp(expr: Expr, env: Env): Value = ???
```

and 2) define the **natural semantics** with **environments**.

$$\sigma \vdash e \Rightarrow n$$

Expressions	$e ::= n$	(Num)
	$e + e$	(Add)
	$e * e$	(Mul)
	$\text{val } x = e; e$	(Val)
	x	(Id)

where

Environments	$\sigma \in \mathbb{X} \xrightarrow{\text{fin}} \mathbb{Z}$	(Env)
Numbers	$n \in \mathbb{Z}$	(BigInt)
Identifiers	$x \in \mathbb{X}$	(String)

```
def interp(expr: Expr, env: Env): Value = expr match
  case Num(n)      => ???
  case Add(l, r)    => ???
  case Mul(l, r)    => ???
  case Val(x, e, b) => ???
  case Id(x)        => ???
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{NUM} \frac{\text{???}}{\sigma \vdash n \Rightarrow \text{???}}$$

$$\text{ADD} \frac{\text{???}}{\sigma \vdash e_1 + e_2 \Rightarrow \text{???}}$$

$$\text{MUL} \frac{\text{???}}{\sigma \vdash e_1 * e_2 \Rightarrow \text{???}}$$

$$\text{VAL} \frac{\text{???}}{\sigma \vdash \text{val } x = e_1; e_2 \Rightarrow \text{???}}$$

$$\text{ID} \frac{\text{???}}{\sigma \vdash x \Rightarrow \text{???}}$$

```
def interp(expr: Expr, env: Env): Value = expr match
  case Num(n)      => ???
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{NUM} \frac{\text{???}}{\sigma \vdash n \Rightarrow \text{???}}$$

```
def interp(expr: Expr, env: Env): Value = expr match
  case Num(n)      => n
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{Num} \frac{}{\sigma \vdash n \Rightarrow n}$$

With the **environment** σ , the **expression** n evaluates to the **number** n .

```
def interp(expr: Expr, env: Env): Value = expr match
  ...
  case Add(l, r)    => ???
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{ADD} \frac{\text{???}}{\sigma \vdash e_1 + e_2 \Rightarrow \text{???}}$$

```
def interp(expr: Expr, env: Env): Value = expr match
...
case Add(l, r)    => interp(l, env) + interp(r, env)
...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{ADD} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma \vdash e_2 \Rightarrow n_2}{\sigma \vdash e_1 + e_2 \Rightarrow n_1 + n_2}$$

With the **environment** σ , the **expression** $e_1 + e_2$ evaluates to the **number** $n_1 + n_2$ when

- With the **environment** σ , the **expression** e_1 evaluates to the **number** n_1 .
- With the **environment** σ , the **expression** e_2 evaluates to the **number** n_2 .

```
def interp(expr: Expr, env: Env): Value = expr match
...
case Mul(l, r)    => interp(l, env) * interp(r, env)
...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{MUL} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma \vdash e_2 \Rightarrow n_2}{\sigma \vdash e_1 * e_2 \Rightarrow n_1 \times n_2}$$

With the **environment** σ , the **expression** $e_1 * e_2$ evaluates to the **number** $n_1 \times n_2$ when

- With the **environment** σ , the **expression** e_1 evaluates to the **number** n_1 .
- With the **environment** σ , the **expression** e_2 evaluates to the **number** n_2 .

```
def interp(expr: Expr, env: Env): Value = expr match
  ...
  case Val(x, e, b) => ???
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{VAL} \frac{\text{???}}{\sigma \vdash \text{val } x = e_1; e_2 \Rightarrow \text{???}}$$

```
def interp(expr: Expr, env: Env): Value = expr match
  ...
  case Val(x, e, b) => ... interp(e, env) ...
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{VAL} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \dots}{\sigma \vdash \text{val } x = e_1; e_2 \Rightarrow ???}$$

With the **environment** σ , the **expression** $\text{val } x = e_1; e_2$ evaluates to the **number** $???$ when

- ① With the **environment** σ , the **expression** e_1 evaluates to the **number** n_1 .
- ② ...

```
def interp(expr: Expr, env: Env): Value = expr match
...
case Val(x, e, b) => ... env + (x -> interp(e, env)) ...
...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{VAL} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma[x \mapsto n_1] \quad \dots}{\sigma \vdash \text{val } x = e_1; e_2 \Rightarrow ???}$$

With the **environment** σ , the **expression** $\text{val } x = e_1; e_2$ evaluates to the **number** $???$ when

- ① With the **environment** σ , the **expression** e_1 evaluates to the **number** n_1 .
- ② With the **environment** $\sigma[x \mapsto n_1], \dots$

```
def interp(expr: Expr, env: Env): Value = expr match
  ...
  case Val(x, e, b) => interp(b, env + (x -> interp(e, env)))
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{VAL} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma[x \mapsto n_1] \vdash e_2 \Rightarrow n_2}{\sigma \vdash \text{val } x = e_1; e_2 \Rightarrow n_2}$$

With the **environment** σ , the **expression** $\text{val } x = e_1; e_2$ evaluates to the **number** n_2 when

- ① With the **environment** σ , the **expression** e_1 evaluates to the **number** n_1 .
- ② With the **environment** $\sigma[x \mapsto n_1]$, the **expression** e_2 evaluates to the **number** n_2 .

```
def interp(expr: Expr, env: Env): Value = expr match
  ...
  case Id(x)           => ???
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{ID} \frac{\text{???}}{\sigma \vdash x \Rightarrow \text{???}}$$

```
def interp(expr: Expr, env: Env): Value = expr match
  ...
  case Id(x)          => env.getOrElse(x, error(s"free identifier: $x"))
  ...
```

$$\sigma \vdash e \Rightarrow n$$

$$\text{ID} \frac{x \in \text{dom}(\sigma)}{\sigma \vdash x \Rightarrow \sigma(x)}$$

With the **environment** σ , the **expression** x evaluates to the **number** $\sigma(x)$ when

- 1 The **variable** x is in the domain of the **environment** σ .

1. Evaluation with Environments

2. Interpreter and Natural Semantics for VAE

Numbers

Addition and Multiplication

Variable Definition

Variable Lookup

3. Examples

Example 1

$$\frac{\text{NUM} \frac{}{\emptyset \vdash 1 \Rightarrow 1} \quad \text{ADD} \frac{\text{ID} \frac{x \in \text{dom}([x \mapsto 1])}{[x \mapsto 1] \vdash x \Rightarrow 1} \quad \text{NUM} \frac{}{[x \mapsto 1] \vdash 2 \Rightarrow 2}}{[x \mapsto 1] \vdash x + 2 \Rightarrow 3}}{\text{VAL} \frac{}{\emptyset \vdash \text{val } x = 1; x + 2 \Rightarrow 3}}$$

Example 1

$$\begin{array}{c}
 \text{NUM} \frac{}{\emptyset \vdash 1 \Rightarrow 1} \quad \text{ID} \frac{x \in \text{dom}([x \mapsto 1])}{[x \mapsto 1] \vdash x \Rightarrow 1} \quad \text{NUM} \frac{}{[x \mapsto 1] \vdash 2 \Rightarrow 2} \\
 \text{VAL} \frac{\text{ADD} \frac{}{[x \mapsto 1] \vdash x + 2 \Rightarrow 3}}{\emptyset \vdash \text{val } x = 1; x + 2 \Rightarrow 3}
 \end{array}$$

We can name environments σ_i to make the derivation tree concise.

$$\begin{array}{c}
 \text{NUM} \frac{}{\emptyset \vdash 1 \Rightarrow 1} \quad \text{ID} \frac{x \in \text{dom}(\sigma_0)}{\sigma_0 \vdash x \Rightarrow 1} \quad \text{NUM} \frac{}{\sigma_0 \vdash 2 \Rightarrow 2} \\
 \text{VAL} \frac{\text{ADD} \frac{}{\sigma_0 \vdash x + 2 \Rightarrow 3}}{\emptyset \vdash \text{val } x = 1; x + 2 \Rightarrow 3}
 \end{array}$$

where

$$\sigma_0 = [x \mapsto 1]$$

Example 2

VAL $\frac{}{\emptyset \vdash \text{val } x = 1; \{\text{val } y = 2; x + y\} \Rightarrow}$

where

$$\begin{array}{c}
 \text{NUM} \frac{}{\emptyset \vdash 1 \Rightarrow 1} \quad \text{VAL} \frac{\text{NUM} \frac{}{\sigma_0 \vdash 2 \Rightarrow 2} \quad \text{ADD} \frac{\text{ID} \frac{x \in \text{dom}(\sigma_1)}{\sigma_1 \vdash x \Rightarrow 1} \quad \text{ID} \frac{y \in \text{dom}(\sigma_1)}{\sigma_1 \vdash y \Rightarrow 2}}{\sigma_1 \vdash x + y \Rightarrow 3}}{\sigma_0 \vdash \text{val } y = 2; x + y \Rightarrow 3}} \\
 \text{VAL} \frac{}{\emptyset \vdash \text{val } x = 1; \{\text{val } y = 2; x + y\} \Rightarrow 3}
 \end{array}$$

where

$$\begin{aligned}
 \sigma_0 &= [x \mapsto 1] \\
 \sigma_1 &= [x \mapsto 1, y \mapsto 2]
 \end{aligned}$$

Example 3

VAL $\frac{}{\emptyset \vdash \text{val } x = 1; \{ \text{val } x = 2; x \} + x \Rightarrow}$

where

$$\begin{array}{c}
 \text{NUM} \frac{}{\emptyset \vdash 1 \Rightarrow 1} \quad \text{VAL} \frac{\text{NUM} \frac{}{\sigma_0 \vdash 2 \Rightarrow 2} \quad \text{ID} \frac{x \in \text{dom}(\sigma_1)}{\sigma_1 \vdash x \Rightarrow 2}}{\sigma_0 \vdash \text{val } x = 2; x \Rightarrow 2} \quad \text{ID} \frac{x \in \text{dom}(\sigma_0)}{\sigma_0 \vdash x \Rightarrow 1} \\
 \text{ADD} \frac{}{\sigma_0 \vdash \{\text{val } x = 2; x\} + x \Rightarrow 3} \\
 \text{VAL} \frac{}{\emptyset \vdash \text{val } x = 1; \{\text{val } x = 2; x\} + x \Rightarrow 3}
 \end{array}$$

where

$$\begin{aligned}
 \sigma_0 &= [x \mapsto 1] \\
 \sigma_1 &= [x \mapsto 2]
 \end{aligned}$$

$$\begin{array}{c}
 \text{NUM} \frac{}{\emptyset \vdash 1 \Rightarrow 1} \quad \text{ID} \frac{x \in \text{dom}(\sigma_0)}{\sigma_0 \vdash x \Rightarrow 1} \\
 \text{VAL} \frac{}{\emptyset \vdash \text{val } x = 1; x \Rightarrow 1} \quad \text{ID} \frac{x \notin \text{dom}(\emptyset)}{\emptyset \vdash x \Rightarrow \text{FAIL}} \\
 \text{ADD} \frac{}{\emptyset \vdash \{\text{val } x = 1; x\} + x \Rightarrow \text{FAIL}}
 \end{array}$$

where

$$\sigma_0 = [x \mapsto 1]$$

We cannot draw the derivation tree for this example because of the **free variable** x in the right-hand side of the addition. The **red** part above is not a real derivation: it marks where **no rule applies**, since the Id rule requires $x \in \text{dom}(\emptyset)$, and **FAIL** merely denotes that the evaluation gets **stuck** there.

```
def interp(expr: Expr, env: Env): Value = expr match
  case Num(n)      => n
  case Add(l, r)    => interp(l, env) + interp(r, env)
  case Mul(l, r)    => interp(l, env) * interp(r, env)
  case Val(x, e, b) => interp(b, env + (x -> interp(e, env)))
  case Id(x)        => env.getOrElse(x, error(s"free identifier: $x"))
```

$$\boxed{\sigma \vdash e \Rightarrow n}$$

$$\text{NUM} \frac{}{\sigma \vdash n \Rightarrow n}$$

$$\text{ADD} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma \vdash e_2 \Rightarrow n_2}{\sigma \vdash e_1 + e_2 \Rightarrow n_1 + n_2}$$

$$\text{MUL} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma \vdash e_2 \Rightarrow n_2}{\sigma \vdash e_1 * e_2 \Rightarrow n_1 \times n_2}$$

$$\text{VAL} \frac{\sigma \vdash e_1 \Rightarrow n_1 \quad \sigma[x \mapsto n_1] \vdash e_2 \Rightarrow n_2}{\sigma \vdash \text{val } x = e_1; e_2 \Rightarrow n_2}$$

$$\text{ID} \frac{x \in \text{dom}(\sigma)}{\sigma \vdash x \Rightarrow \sigma(x)}$$

<https://github.com/ku-plrg-classroom/docs/tree/main/cose212/vae>

- Please see above document on GitHub:
 - Implement `interp` function.
 - Implement `freeIds` function.
 - Implement `bindingIds` function.
 - Implement `boundIds` function.
 - Implement `shadowedIds` function.
- It is just an exercise, and you **don't need to submit** anything.
- However, some exam questions might be related to this exercise, so we **strongly recommend** solving it by yourself.

- First-Order Functions

Jihyeok Park

`jihyeok_park@korea.ac.kr`

`https://plrg.korea.ac.kr`